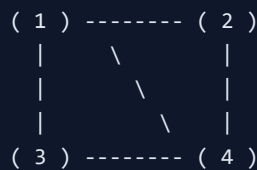


Graph Introduction & Terminologies — Data Structures

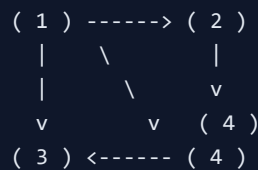
Definition: A **Graph** is a non-linear data structure denoted as $G = (V, E)$, consisting of a non-empty finite set of **Vertices (Nodes) V** and a set of **Edges (Arcs) E** connecting pairs of vertices.

1. Detailed Technical Explanation

UNDIRECTED GRAPH:



DIRECTED GRAPH (DIGRAPH):



Core Graph Terminologies:

1. **Vertex (Node):** An individual data point or entity in the graph.
2. **Edge (Arc):** A link or line connecting two vertices (u, v) .
3. **Directed Graph (Digraph):** A graph where edges have a defined direction (ordered pair (u, v) or (v, u)).
4. **Undirected Graph:** A graph where edges are bidirectional and symmetric $((u, v) = (v, u))$.
5. **Weighted Graph:** A graph where each edge is assigned a numerical cost, distance, or weight $w(u, v)$.
6. **Degree of a Vertex:**
 - In **Undirected Graphs**: The total number of edges connected to that vertex.
 - In **Directed Graphs**:
 - **In-Degree**: Number of edges directed *into* the vertex.
 - **Out-Degree**: Number of edges directed *out of* the vertex.
7. **Handshaking Lemma:** In any undirected graph, the sum of degrees of all vertices equals twice the number of edges:

$$\text{Sum of Deg}(v) = 2 * |E|$$
8. **Adjacent Vertices (Neighbors):** Two vertices connected directly by an edge.
9. **Path:** A sequence of alternating vertices and edges connecting a source vertex to a destination vertex.
10. **Cycle:** A closed path where the start vertex and end vertex are the same, with no edge repeated.
11. **Acyclic Graph:** A graph containing zero cycles (e.g., Trees, Directed Acyclic Graphs - DAG).
12. **Connected Graph (Undirected):** A graph where there exists at least one path between every pair of vertices.
13. **Strongly Connected Graph (Directed):** A digraph where there is a directed path from every vertex to every other vertex.
14. **Complete Graph (Kn):** A graph where every vertex is connected to every other vertex. Number of edges in K_n :

$$|E| = n * (n - 1) / 2$$

2. Memory Keywords & Properties

- **Vertices & Edges:** $G = (V, E)$.
 - **Handshaking Lemma:** $\sum \deg(v) = 2|E|$ (Total odd degree vertices is always EVEN).
 - **Dense vs Sparse Graph:** Dense when $|E| \approx |V|^2$; Sparse when $|E| \approx |V|$.
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3. Must-Write Points for Exams

- In an undirected complete graph with N vertices, the maximum number of edges is $N(N - 1) / 2$.
 - In a directed complete graph with N vertices, the maximum number of edges is $N(N - 1)$.
 - Trees are a special type of connected, acyclic undirected graph with $N - 1$ edges.
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4. Quick Recall Flow

$G = (V, E) \rightarrow$ Directed vs Undirected $\rightarrow$ In/Out Degree $\rightarrow$ Handshaking Lemma ($\sum \text{Deg} = 2E$) $\rightarrow$ Complete Graph $E = N(N-1)/2$